

Viscosity Solutions and its Applications to Optimal Stopping

1 The Finite-Horizon Control Problem

We now consider the stochastic control problem over a finite time interval $[0, T]$, where $T \in (0, \infty)$ is the fixed terminal time. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space equipped with a filtration $\{\mathcal{F}_t\}$ and a d -dimensional standard Brownian motion $W = \{W_t\}_{t \geq 0}$.

Starting from state $x \in \mathbb{R}^n$ at time $t \in [0, T]$, the state process X with state space $S = \mathbb{R}^n$ evolves for $s \in [t, T]$ according to the controlled stochastic differential equation (SDE):

$$dX_s = b(s, X_s, U_s)ds + \sigma(s, X_s, U_s)dW_s, \quad X_t = x. \quad (1)$$

The control space U is a subset of $\mathbb{R}^p$. The functions $b : [0, T] \times \mathbb{R}^n \times U \rightarrow \mathbb{R}^n$ and $\sigma : [0, T] \times \mathbb{R}^n \times U \rightarrow \mathbb{R}^{n \times d}$ govern the plant dynamics.

1.1 Rigorous Markovian and Time Properties

To apply dynamic programming rigorously, we must address the structural nature of the control U_s . If U_s is merely an open-loop adapted process, it can depend arbitrarily on the history of the Brownian motion. Consequently, the solution X_s to (1) would be an Itô process, but generally *not* a Markov process, shattering the memoryless property required for recursive optimization.

Therefore, we restrict our set of admissible controls, denoted $\mathcal{A}(t)$, to **Markovian feedback controls**. A control is admissible if there exists a deterministic, Borel measurable function $u : [0, T] \times \mathbb{R}^n \rightarrow U$ such that $U_s = u(s, X_s)$. Under this restriction, X_s becomes a rigorous Markov process.

Furthermore, the finite-horizon problem is strictly **time-inhomogeneous**. This manifests because the plant coefficients and the running cost depend explicitly on the absolute time s , and the presence of a hard stopping horizon T breaks time-translation invariance. Consequently, the transition probability kernel depends on both the starting and ending times, and the value function must explicitly be a function of both time and space: $V = V(t, x)$.

The cost-to-go function for using an admissible control process $U \in \mathcal{A}(t)$ starting from initial state x at time t is:

$$J(t, x, U) = \mathbb{E} \left[\int_t^T g(s, X_s, U_s)ds + h(X_T) \right], \quad (2)$$

where g is the running cost and h is the terminal cost. The value function $V : [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}$ is defined as:

$$V(t, x) = \inf_{U \in \mathcal{A}(t)} J(t, x, U). \quad (3)$$

Assumption 1.1. Throughout this chapter, we assume:

- (a) The control space U is a compact subset of $\mathbb{R}^p$.
- (b) The functions b, σ, g , and h are continuous.
- (c) There exists a constant $L > 0$ such that for all $s \in [0, T]$, $x, y \in \mathbb{R}^n$, and $\alpha \in U$, the functions b, σ, g , and h satisfy a uniform Lipschitz condition in the spatial variable.
- (d) The functions satisfy a linear growth condition: $|b| + \|\sigma\| + |g| + |h| \leq L(1 + |x|)$.

2 Dynamic Programming and the Finite-Horizon HJB

We state the Dynamic Programming Principle (DPP) for the finite-horizon problem.

Theorem 2.1 (Dynamic Programming Principle). *Let $t \in [0, T]$ and $x \in \mathbb{R}^n$, and suppose Assumption 1.1 holds. For any stopping time τ taking values in $[t, T]$, we have:*

$$V(t, x) = \inf_{U \in \mathcal{A}(t)} \mathbb{E} \left[\int_t^\tau g(s, X_s^U, U_s) ds + V(\tau, X_\tau^U) \right], \quad (4)$$

where X^U denotes the state process corresponding to (t, x, U) .

Sketch of Proof. We denote the RHS of (4) by $\bar{V}(t, x)$. Let $0 \leq t \leq \tau \leq T$, $x \in \mathbb{R}^n$, and $U \in \mathcal{A}(t)$, and let X^U be the state process corresponding to (t, x, U) . By the tower property of conditional expectation and the definition of the cost function, we have:

$$\begin{aligned} J(t, x, U) &= \mathbb{E} \left[\int_t^\tau g(s, X_s^U, U_s) ds \right] + \mathbb{E} \left[\mathbb{E} \left[\int_\tau^T g(s, X_s^U, U_s) ds + h(X_T^U) \mid \mathcal{F}_\tau \right] \right] \\ &= \mathbb{E} \left[\int_t^\tau g(s, X_s^U, U_s) ds \right] + \mathbb{E}[J(\tau, X_\tau^U, U)] \\ &\geq \mathbb{E} \left[\int_t^\tau g(s, X_s^U, U_s) ds \right] + \mathbb{E}[V(\tau, X_\tau^U)]. \end{aligned}$$

Taking the infimum over $U \in \mathcal{A}(t)$ yields $V(t, x) \geq \bar{V}(t, x)$.

We now consider the reverse inequality. Let $\varepsilon > 0$ and let $U_\varepsilon \in \mathcal{A}(\tau)$ be an ε -optimal control such that:

$$V(\tau, X_\tau^U) \leq J(\tau, X_\tau^U, U_\varepsilon) \leq V(\tau, X_\tau^U) + \varepsilon.$$

Define a concatenated process $\bar{U}$ on $[t, T]$ by:

$$\bar{U}(s) = \begin{cases} U(s) & t \leq s < \tau, \\ U_\varepsilon(s) & \tau \leq s \leq T. \end{cases}$$

Assuming $\bar{U} \in \mathcal{A}(t)$, let $\bar{X}$ be the state process corresponding to $(t, x, \bar{U})$. Then:

$$\begin{aligned} V(t, x) &\leq J(t, x, \bar{U}) = \mathbb{E} \left[\int_t^T g(s, \bar{X}_s, \bar{U}_s) ds + h(\bar{X}_T) \right] \\ &= \mathbb{E} \left[\int_t^\tau g(s, X_s^U, U_s) ds \right] + \mathbb{E}[J(\tau, X_\tau^U, U_\varepsilon)] \\ &\leq \mathbb{E} \left[\int_t^\tau g(s, X_s^U, U_s) ds \right] + \mathbb{E}[V(\tau, X_\tau^U)] + \varepsilon. \end{aligned}$$

Taking the infimum over $U \in \mathcal{A}(t)$ on both sides yields $V(t, x) \leq \bar{V}(t, x) + \varepsilon$. Since $\varepsilon > 0$ is arbitrary, it follows that $V(t, x) \leq \bar{V}(t, x)$, completing the sketch. $\square$

Remark 2.2. *A rigorous proof of the Dynamic Programming Principle necessitates the weak formulation of the stochastic control problem, which carefully handles the measurability and admissibility of the concatenated control strategy $\bar{U}$. We refer the reader to Yong and Zhou (1999) for a complete measure-theoretic proof, which relies fundamentally on the assumption that the underlying filtration is the natural filtration generated by the Brownian motion. Alternatively, Fleming and Soner (2006) provide a proof utilizing a discrete-time approximation argument.*

If we assume classical smoothness, we can rigorously derive the Hamilton-Jacobi-Bellman (HJB) equation directly from the DPP. Assume $V \in C^{1,2}([0, T] \times \mathbb{R}^n)$ and that an optimal Markov control policy $u^*(t, x)$ exists.

For an arbitrary increment $h > 0$, we apply Itô's Lemma to the process $V(s, X_s)$ from $s = t$ to $s = t + h$ using an arbitrary constant control $\alpha \in U$:

$$\mathbb{E}[V(t + h, X_{t+h})] = V(t, x) + \mathbb{E} \int_t^{t+h} \left(\frac{\partial V}{\partial s}(s, X_s) + A^\alpha V(s, X_s) \right) ds,$$

where the generator A^α is given by:

$$A^\alpha \phi(s, x) = b(s, x, \alpha) \cdot \nabla \phi(s, x) + \frac{1}{2} \text{Tr} \left(\sigma(s, x, \alpha) \sigma(s, x, \alpha)^T \nabla^2 \phi(s, x) \right). \quad (5)$$

Substituting this expectation back into the DPP (4), the $V(t, x)$ terms cancel, yielding:

$$0 \leq \mathbb{E} \left[\int_t^{t+h} \left(\frac{\partial V}{\partial s}(s, X_s) + A^\alpha V(s, X_s) + g(s, X_s, \alpha) \right) ds \right].$$

Dividing by h and taking the limit as $h \downarrow 0$, the Dominated Convergence Theorem and the continuity of the paths yield the local differential inequality:

$$0 \leq \frac{\partial V}{\partial t}(t, x) + A^\alpha V(t, x) + g(t, x, \alpha), \quad \forall \alpha \in U.$$

Because the optimal control u^* achieves strict equality, we may take the infimum over all possible actions $\alpha \in U$ to consolidate the conditions into the finite-horizon HJB equation:

$$0 = \frac{\partial V}{\partial t}(t, x) + \inf_{\alpha \in U} [A^\alpha V(t, x) + g(t, x, \alpha)], \quad (t, x) \in [0, T] \times \mathbb{R}^n, \quad (6)$$

with the terminal Cauchy boundary condition $V(T, x) = h(x)$.

3 Continuity of the Value Function

Before introducing viscosity solutions, we must establish that the value function possesses the baseline topological regularity required to be a candidate solution (specifically, $V \in C(Q)$). The following theorem shows that the underlying structure of the controlled SDE is inherited by the value function, guaranteeing it is Lipschitz continuous in the state variable and Hölder continuous in the time variable.

Theorem 3.1 (Continuity of the Value Function). *Under Assumption 1.1, the value function $V(t, x)$ is continuous on $[0, T] \times \mathbb{R}^n$. Moreover, there exists a constant $K > 0$ such that for all $t, s \in [0, T]$ and $x, y \in \mathbb{R}^n$:*

$$|V(t, x)| \leq K(1 + |x|), \quad (7)$$

$$|V(t, x) - V(s, y)| \leq K(|x - y| + (1 + \max(|x|, |y|))|t - s|^{1/2}). \quad (8)$$

4 Viscosity Solutions of the HJB Equation

The major obstacle to the verification procedure is that it requires the value function to be a classical $C^{1,2}$ solution. However, in stochastic control, optimal paths often switch abruptly, creating sharp "kinks" in the value function where classical derivatives cease to exist. To circumvent this, we utilize the theory of viscosity solutions, which transfers the burden of differentiation from the non-smooth value function onto smooth "test functions."

To write the HJB equation in the standard literature form, we introduce the Hamiltonian H . Let $\mathbb{S}_+^n$ denote the set of symmetric, nonnegative definite $n \times n$ matrices. We define $H : [0, T) \times \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{S}_+^n \rightarrow \mathbb{R}$ by:

$$H(s, z, p, M) = \sup_{\alpha \in U} \left[-p \cdot b(s, z, \alpha) - \frac{1}{2} \text{Tr}(Ma(s, z, \alpha)) - g(s, z, \alpha) \right], \quad (9)$$

where $a = \sigma \sigma^T$.

We can rewrite the HJB equation (6) using H . Exploiting the fundamental algebraic identity $\inf(f) = -\sup(-f)$, we multiply (6) by -1 :

$$0 = -\frac{\partial V}{\partial t}(t, x) + \sup_{\alpha \in U} [-A^\alpha V(t, x) - g(t, x, \alpha)].$$

Substituting $p = \nabla V(t, x)$ and $M = \nabla^2 V(t, x)$ into the Hamiltonian, the HJB equation elegantly becomes:

$$-\frac{\partial V}{\partial t}(t, x) + H(t, x, \nabla V(t, x), \nabla^2 V(t, x)) = 0. \quad (10)$$

Definition 4.1 (Viscosity Solution). *Let $Q = [0, T) \times \mathbb{R}^n$ and suppose $V \in C(Q)$.*

1. *V is a **viscosity subsolution** of (10) if, for all $(s, z) \in Q$ and all $\phi \in C^{1,2}(Q)$ such that $V - \phi$ has a local maximum at (s, z) , we have:*

$$-\frac{\partial \phi}{\partial t}(s, z) + H(s, z, \nabla \phi(s, z), \nabla^2 \phi(s, z)) \leq 0.$$

2. *V is a **viscosity supersolution** of (10) if, for all $(s, z) \in Q$ and all $\phi \in C^{1,2}(Q)$ such that $V - \phi$ has a local minimum at (s, z) , we have:*

$$-\frac{\partial \phi}{\partial t}(s, z) + H(s, z, \nabla \phi(s, z), \nabla^2 \phi(s, z)) \geq 0.$$

3. *V is a **viscosity solution** if it is both a viscosity subsolution and supersolution.*

5 Proving the Value Function is a Viscosity Solution

We now prove rigorously that the value function satisfies these properties.

Lemma 5.1. *Under Assumption 1.1, the value function V is a viscosity subsolution of (10).*

Proof. Fix $(s, z) \in Q$ and let $\phi \in C^{1,2}(Q)$ be a test function such that $V - \phi$ has a global maximum at (s, z) . We may assume without loss of generality that $(V - \phi)(s, z) = 0$. (If the maximum value were a constant C , we would simply define a new test function $\tilde{\phi} = \phi + C$. This vertical shift preserves all derivatives of ϕ while perfectly aligning $\tilde{\phi}(s, z) = V(s, z)$). Thus, $V(t, x) \leq \phi(t, x)$ globally.

For an arbitrary $\alpha \in U$, consider the constant open-loop control $U_r \equiv \alpha$. Let X be the resulting state process. Fix $\rho > 0$ and $h \in (0, T - s)$, and define the stopping time $\tau_h = (s + h) \wedge \inf\{r \geq s : |X_r - z| \geq \rho\}$. Applying Itô's rule to $\phi(\tau_h, X_{\tau_h})$ and taking expectations:

$$\mathbb{E}[\phi(\tau_h, X_{\tau_h})] = \phi(s, z) + \mathbb{E} \int_s^{\tau_h} \left[\frac{\partial \phi}{\partial t}(r, X_r) + A^\alpha \phi(r, X_r) \right] dr. \quad (11)$$

By the DPP, $V(s, z) \leq \mathbb{E} \left[\int_s^{\tau_h} g(r, X_r, \alpha) dr + V(\tau_h, X_{\tau_h}) \right]$. Since $V(s, z) = \phi(s, z)$ and $V \leq \phi$, we have:

$$\phi(s, z) \leq \mathbb{E} \left[\int_s^{\tau_h} g(r, X_r, \alpha) dr + \phi(\tau_h, X_{\tau_h}) \right]. \quad (12)$$

Combining (11) and (12), dividing by h , and letting $h \rightarrow 0$:

$$0 \leq \frac{\partial \phi}{\partial t}(s, z) + A^\alpha \phi(s, z) + g(s, z, \alpha). \quad (13)$$

Crucially, this inequality holds for *any* choice of $\alpha \in U$. Therefore, it must remain true even if we select the α that minimizes the right-hand side. Taking the infimum over $\alpha \in U$ yields:

$$0 \leq \frac{\partial \phi}{\partial t}(s, z) + \inf_{\alpha \in U} [A^\alpha \phi(s, z) + g(s, z, \alpha)]. \quad (14)$$

Multiplying by -1 transforms the infimum into the supremum Hamiltonian H , resulting precisely in the subsolution inequality. $\square$

Lemma 5.2. *Under Assumption 1.1, the value function V is a viscosity supersolution of (10).*

Proof. Fix $(s, z) \in Q$ and let $\phi \in C^{1,2}(Q)$ such that $V - \phi$ has a global minimum at (s, z) . Without loss of generality, let $V(s, z) = \phi(s, z)$, so $V \geq \phi$.

Fix $\varepsilon > 0$. By the definition of the infimum, there exists an ε -optimal control $U^{\varepsilon, h} \in \mathcal{A}(s)$ such that for the localized stopping time $\tau_{\varepsilon, h}$:

$$\mathbb{E} \left[\int_s^{\tau_{\varepsilon, h}} g(r, X_r, U_r^{\varepsilon, h}) dr + V(\tau_{\varepsilon, h}, X_{\tau_{\varepsilon, h}}) \right] \leq V(s, z) + \varepsilon h. \quad (15)$$

Applying Itô's rule to ϕ and substituting $V(s, z) = \phi(s, z)$ and $V \geq \phi$, we obtain:

$$\mathbb{E} \int_s^{\tau_{\varepsilon, h}} \left[-\frac{\partial \phi}{\partial t}(r, X_r) - A^{U_r^{\varepsilon, h}} \phi(r, X_r) - g(r, X_r, U_r^{\varepsilon, h}) \right] dr \geq -\varepsilon h. \quad (16)$$

To eliminate the specific dependency on the ε -optimal control $U_r^{\varepsilon, h}$, we actively bound the integrand from below. We replace the terms depending on $U_r^{\varepsilon, h}$ with the supremum taken over all possible actions $\alpha \in U$. This mathematically guarantees the inequality is maintained:

$$\mathbb{E} \int_s^{\tau_{\varepsilon, h}} \left[-\frac{\partial \phi}{\partial t}(r, X_r) + \sup_{\alpha \in U} (-A^\alpha \phi(r, X_r) - g(r, X_r, \alpha)) \right] dr \geq -\varepsilon h. \quad (17)$$

Dividing by h and letting $h \rightarrow 0$, the limit yields:

$$-\frac{\partial \phi}{\partial t}(s, z) + H(s, z, \nabla \phi(s, z), \nabla^2 \phi(s, z)) \geq -\varepsilon. \quad (18)$$

Since $\varepsilon > 0$ was arbitrary, the supersolution property follows. $\square$

Theorem 5.3. *Under Assumption 1.1, the value function V is the unique viscosity solution of (10) satisfying the terminal condition $V(T, x) = h(x)$.*

6 Optimal Stopping via Viscosity Solutions

We now apply the viscosity framework to optimal stopping problems, allowing us to completely bypass classical C^2 requirements and standard Verification Theorems.

Consider a time-homogeneous uncontrolled diffusion $dX_t = b(X_t)dt + \sigma(X_t)dW_t$. We seek to minimize the cost functional:

$$J(x, \tau) = \mathbb{E} \left[\int_0^\tau e^{-\gamma s} g(X_s) ds + e^{-\gamma \tau} h(X_\tau) \mathbf{1}_{\{\tau < \infty\}} \right], \quad (19)$$

where the value function is $V(x) = \inf_{\tau \in \mathcal{S}} J(x, \tau)$. The Variational Inequality (VI) associated with this problem is:

$$\min\{-\gamma V(x) + AV(x) + g(x), h(x) - V(x)\} = 0, \quad x \in \mathbb{R}^n. \quad (20)$$

Definition 6.1 (Viscosity Solution for Optimal Stopping). *Let $V \in C^0(\mathbb{R}^n)$.*

1. V is a **viscosity subsolution** if, for every $x_0 \in \mathbb{R}^n$ and every test function $\phi \in C^2(\mathbb{R}^n)$ such that $V - \phi$ attains a local maximum at x_0 with $V(x_0) = \phi(x_0)$:

$$\min\{-\gamma V(x_0) + A\phi(x_0) + g(x_0), h(x_0) - V(x_0)\} \geq 0.$$

2. V is a **viscosity supersolution** if, for every $x_0 \in \mathbb{R}^n$ and every test function $\phi \in C^2(\mathbb{R}^n)$ such that $V - \phi$ attains a local minimum at x_0 with $V(x_0) = \phi(x_0)$:

$$\min\{-\gamma V(x_0) + A\phi(x_0) + g(x_0), h(x_0) - V(x_0)\} \leq 0.$$

To prove these properties, we rely on the **Weak Dynamic Programming Principle**, which holds unconditionally for any arbitrary stopping time $\theta \in \mathcal{S}$:

$$V(x) = \inf_{\tau \in \mathcal{S}} \mathbb{E} \left[\int_0^{\tau \wedge \theta} e^{-\gamma s} g(X_s) ds + e^{-\gamma(\tau \wedge \theta)} (h(X_\tau) \mathbf{1}_{\{\tau < \theta\}} + V(X_\theta) \mathbf{1}_{\{\theta \leq \tau\}}) \right]. \quad (21)$$

Theorem 6.2. *Assuming $V \in C^0(\mathbb{R}^n)$, the value function $V(x)$ is a viscosity subsolution of the VI (20).*

Proof. Fix $x_0 \in \mathbb{R}^n$ and a test function $\phi \in C^2$ such that $V - \phi$ has a local maximum at x_0 with $V(x_0) = \phi(x_0)$. There exists $R > 0$ such that $V(x) \leq \phi(x)$ on $B_R(x_0)$. First, evaluating the immediate stopping strategy $\tau = 0$ yields $V(x_0) \leq h(x_0)$, establishing $h(x_0) - V(x_0) \geq 0$. Second, let τ_R be the exit time from $B_R(x_0)$. Fix $t > 0$ and let

the horizon $\theta = t \wedge \tau_R$. Under the sub-optimal strategy of never voluntarily stopping ($\tau = \infty$), the Weak DPP (21) gives:

$$V(x_0) \leq \mathbb{E} \left[\int_0^\theta e^{-\gamma s} g(X_s) ds + e^{-\gamma \theta} V(X_\theta) \right].$$

Since $X_\theta \in B_R(x_0)$, $V(X_\theta) \leq \phi(X_\theta)$. Applying Itô's formula to $e^{-\gamma s} \phi(X_s)$ up to θ :

$$V(x_0) \leq \mathbb{E} \left[\int_0^\theta e^{-\gamma s} g(X_s) ds \right] + \phi(x_0) + \mathbb{E} \left[\int_0^\theta e^{-\gamma s} (-\gamma \phi(X_s) + A\phi(X_s)) ds \right].$$

Since $V(x_0) = \phi(x_0)$, the terms cancel. Dividing by t and letting $t \downarrow 0$, the limits yield $0 \leq -\gamma V(x_0) + A\phi(x_0) + g(x_0)$. Because both components are non-negative, their minimum is non-negative, completing the subsolution proof. $\square$

Theorem 6.3. *Assuming $V \in C^0(\mathbb{R}^n)$, the value function $V(x)$ is a viscosity supersolution of the VI (20).*

Proof. Fix $x_0 \in \mathbb{R}^n$ and $\phi \in C^2$ such that $V - \phi$ attains a local minimum at x_0 with $V(x_0) = \phi(x_0)$. There exists $R > 0$ such that $V \geq \phi$ on $B_R(x_0)$. We proceed by contradiction. Assume:

$$\min\{-\gamma V(x_0) + A\phi(x_0) + g(x_0), h(x_0) - V(x_0)\} > 0.$$

This implies $h(x_0) - V(x_0) > \delta > 0$ and $-\gamma \phi(x_0) + A\phi(x_0) + g(x_0) \geq c > 0$. By continuity, we can find a smaller ball $B_r(x_0)$ where these strict bounds hold uniformly. Let τ_r be the exit time from B_r , fix $t > 0$, and define $\theta = t \wedge \tau_r$.

For any $\varepsilon > 0$, there exists an ε -optimal stopping time τ^ε . Using the Weak DPP evaluated at horizon θ :

$$V(x_0) + \varepsilon \geq \mathbb{E} \left[\int_0^{\tau^\varepsilon \wedge \theta} e^{-\gamma s} g(X_s) ds + e^{-\gamma(\tau^\varepsilon \wedge \theta)} (h(X_{\tau^\varepsilon}) \mathbf{1}_{\{\tau^\varepsilon < \theta\}} + V(X_\theta) \mathbf{1}_{\{\theta \leq \tau^\varepsilon\}}) \right].$$

Inside $B_r(x_0)$, $h(X_{\tau^\varepsilon}) \geq \phi(X_{\tau^\varepsilon})$ and $V(X_\theta) \geq \phi(X_\theta)$. Replacing both payloads with ϕ , we apply Itô's formula to $e^{-\gamma s} \phi(X_s)$. Canceling $V(x_0) = \phi(x_0)$ yields:

$$\varepsilon \geq \mathbb{E} \left[\int_0^{\tau^\varepsilon \wedge \theta} e^{-\gamma s} (-\gamma \phi(X_s) + A\phi(X_s) + g(X_s)) ds \right].$$

Because the integrand is strictly bounded below by $c > 0$ inside $B_r(x_0)$, we have:

$$\varepsilon \geq c \mathbb{E} \left[\int_0^{\tau^\varepsilon \wedge \theta} e^{-\gamma s} ds \right].$$

Since $h(x_0) > V(x_0)$ strictly, the ε -optimal strategy cannot stop immediately. The process must evolve for a strictly positive time almost surely, meaning the integral evaluates to a fixed positive constant $K > 0$. We arrive at $\varepsilon \geq cK$. Choosing an $\varepsilon < cK$ yields a direct mathematical contradiction. Thus, the minimum must be less than or equal to zero. $\square$